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Detecting network activity from sampled network metadata

Abstract

Generally discussed herein are devices, systems, and methods for cloud traffic monitoring. A method can include receiving sampled network metadata of a packet transmitted via a computer network, providing the sampled network metadata to a neural network (NN) trained on labeled sampled network metadata, and providing, based on only the sampled network metadata, a classification for the sampled network metadata via the trained neural network.

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Background/Summary

BACKGROUND

(1) If an organization wants to provide access to secure cloud applications, it can monitor deep packet information. Monitoring deep packet information includes inspecting an entire packet, including the header and payload. The deep packet information is different from conventional packet filtering which examines only packet headers. Deep packet inspection is typically performed as a firewall feature at the application layer of the Open Systems Interconnection (OSI) reference model. Deep packet inspection can be used to detect and intercept viruses and other forms of malicious traffic. Deep packet inspection has privacy issues in that an originator and a recipient of content can be identified using deep packet inspection. The existence of secure socket layer (SSL) interception, in which encrypted traffic is intercepted, decrypted, and analyzed, only increases these concerns.

(2) Deep packet information can be overwhelmingly large and it can be difficult to sift through all deep packet information or even a subset of the packets of deep packet information. Further, deep packet inspection can create new vulnerabilities, such as buffer overflow attacks, denial-of-service (DOS) attacks, and some types of malware. Further yet, deep packet inspection software requires its own periodic updates and revisions to remain optimally effective. Even further yet, deep packet inspection reduces network speed, because it increases the burden on network processors.

(3) In a cloud setting, a provider generally does not like to give a customer access to network traffic within the multi-customer platform because of the risk of customer-to-customer data transfer or data leaks. The cloud provider can effectively encapsulate the deep packet information to separate the customer and management data flows. This often means that deep packet inspection technologies can have trouble processing cloud traffic as the deep packet inspection technologies are not typically privy to the encapsulation techniques and how to de-encapsulate the packets.

SUMMARY

(4) This summary section is provided to introduce aspects of embodiments in a simplified form, with further explanation of the embodiments following in the detailed description. This summary section is not intended to identify essential or required features of the claimed subject matter, and the combination and order of elements listed in this summary section are not intended to provide limitation to the elements of the claimed subject matter.

(5) Systems, methods, device, and computer or other machine-readable media can provide improvements over prior cloud provisioning solutions. The improvements can include classifying network behavior based on sampled network metadata (only sampled network metadata). The improvements can include reducing memory and compute bandwidth overhead required as compared to prior network behavior monitoring techniques. The improvements can further include reduction of man-in-the-middle time used to identify malignant network behavior.

(6) A method, device, computer-readable medium, a means for, and system for network behavior monitoring are provided. The method, device, computer-readable medium, means for, or system can be configured to receive sampled network metadata of a packet transmitted via a computer network, provide the sampled network metadata to a neural network (NN) trained on labeled sampled network metadata, and provide, based on only the sampled network metadata, a classification for the sampled network metadata via the trained neural network.

(7) The device can include processing circuitry, a memory including instructions that when executed by the processing circuitry cause the processing circuitry to perform operations, or other circuitry. The device can include a router, switch, firewall, or client device.

(8) The NN can be trained further based on contents of the packet and the label is an actual classification associated with the contents of the packet and associated sampled network metadata.

The actual classification can be determined using deep packet inspection. The NN can be a recurrent NN. The NN can include a bi-directional long short term memory (LSTM) NN.

(9) The sampled network metadata can from network traffic provided over layer three of the computer network. The actual classification can include one or more of a user authentication, a device authentication, a database query, file transfer, data streaming, or a malicious action.

Description

BRIEF DESCRIPTION OF DRAWINGS

- (1) FIG. 1 illustrates, by way of example, a diagram of an embodiment of a cloud resource provisioning system.
- (2) FIG. 2 illustrates, by way of example, a diagram of an embodiment of a system for generating sampled network metadata.
- (3) FIG. 3 illustrates, by way of example, a conceptual block diagram of a system for training a neural network (NN) to classify based on the sampled metadata.
- (4) FIG. 4 illustrates, by way of example, a conceptual block diagram of an embodiment of a system for using the NN after training.
- (5) FIG. 5 illustrates, by way of example, a conceptual block diagram of an embodiment of computer network traffic analysis system.
- (6) FIG. 6 illustrates, by way of example, a conceptual block diagram of an embodiment of an LSTM NN.
- (7) FIG. 7 illustrates, by way of example, a flow diagram of an embodiment of a method for cloud security.
- (8) FIG. 8 illustrates, by way of example, a block diagram of an embodiment of a machine (e.g., a computer system) to implement one or more embodiments.

DETAILED DESCRIPTION

(9) In the following description, reference is made to the accompanying drawings that form a part hereof, and in which is shown by way of illustration specific embodiments which may be practiced. These embodiments are described in sufficient detail to enable those skilled in the art to practice the embodiments. It is to be understood that other embodiments may be utilized and that structural, logical, and/or electrical changes may be made without departing from the scope of the embodiments. The following description of embodiments is, therefore, not to be taken in a limited sense, and the scope of the embodiments is defined by the appended claims.

(10) The operations, functions, or techniques described herein may be implemented in software in some embodiments. The software may include computer executable instructions stored on computer or other machine-readable media or storage device, such as one or more non-transitory memories (e.g., a non-transitory machine-readable medium) or other type of hardware-based storage devices, either local or networked. Further, such functions may correspond to subsystems, which may be software, hardware, firmware or a combination thereof. Multiple functions may be performed in one or more subsystems as desired, and the embodiments described are merely examples. The software may be executed on a digital signal processor, application specific integrated circuitry (ASIC), microprocessor, central processing unit (CPU), graphics processing unit (GPU), field programmable gate array (FPGA), or other type of processor operating on a computer system, such as a personal computer, server or other computer system, turning such computer system into a specifically programmed machine. The functions, operations, or methods may be implemented using processing circuitry, such as may include electric and/or electronic components (e.g., one or more transistors, resistors, capacitors, inductors, amplifiers, modulators, demodulators, antennas, radios, regulators, diodes, oscillators, multiplexers, logic gates (e.g., AND, OR, XOR, negate, or the like), buffers, caches, memories, GPUs, CPUs, FPGAs, ASICs, or the

like).

(11) It is quite difficult to provide network security in the cloud. It is even more difficult to preserve privacy, maintain network bandwidth, and not consume too many compute resources (e.g., firewall, gateway, router, switch, or the like), while providing network security. The network security is often provided using deep packet inspection, which neither provides privacy, maintains network bandwidth, nor consumes a small number of compute resources, as previously discussed in the background.

(12) Cloud providers have yet to develop an efficient technique for the detection of malicious activity that can scale to millions of customers. One way to do this is to make use of network data, since most attacks have a network component, usually at multiple stages of the kill chain. However, the detection of malicious activity over network data is challenging, since network data consumes a large volume of memory and analysis of the large volume of data is time and resource intensive.

(13) Sampled network metadata, discussed in more detail elsewhere, is typically useful for detection of volumetric attacks with a clear metadata signature (such as a SYN flooding denial of service (DoS) attack). Other attacks, which involve a volumetric component, have a less apparent metadata signature. For example, the transfer of a payload, or data exfiltration over some network protocol, may be observable in sampled metadata, but it is unclear how to distinguish these payload transfer or data exfiltration from other traffic.

(14) Embodiments provide a supervised-learning based approach for detecting network features that typically require deep packet information (such as whether a file transfer or streaming occurred) from sampled network metadata. The approach is based on training a recurrent machine learning (ML) model (for example, a bidirectional LSTM) that translates between sampled network metadata, and classifies based on the corresponding network traffic. This can be done on publicly available datasets that include both network metadata and corresponding deep packet information. The model can then be used to detect network features that are typically only evident from deep packet inspection but based on only sampled network metadata.

(15) The model can be useful beyond the security domain and can be extended to other domains that rely on sampled network metadata. One such application is to monitor resource usage patterns (e.g., file transfers, streaming, etc.) from sampled network activity, without installing an agent and without deep packet inspection. Embodiments can provide such efficient monitoring at scale.

(16) Inferring network activity patterns from sampled network metadata is a difficult problem that can be beneficial for security applications. A model can be trained to identify normal (“benign”) patterns of network activity. These patterns are abundant and robust in publicly available datasets so the trained model is able to robustly classify the normal patterns. The model can then be used to translate between the sampled metadata and the actual underlying network activity patterns, making it applicable to various classification, monitoring, and detection tasks.

(17) Embodiments can provide advantages, such as: 1) providing advantages of deep packet inspection without requiring all information of the deep packets, 2) performing operations of deep packet inspection without compromising privacy, 3) performing operations of deep packet inspection with fewer compute resources. Embodiments can provide these advantages by training a neural network (NN) based on deep packet inspection information, sampled metadata, and corresponding classifications determined based on the deep packet inspection. The NN can then be deployed to operate on only the sampled metadata to identify network operations.

(18) FIG. 1 illustrates, by way of example, a diagram of an embodiment of a cloud application system **100**. The system **100** provides a user **101** with access to functionality of applications, hardware, or data hosted remote to the user **101**, in “the cloud” **106**. The user **101** can access the functionality or data of the cloud **106** using a device **102**. The device **102** can include a user interface **116**. The user interface **116** provides the user **101** with access to functionality of the device **102**. A web browser **120**, accessible through the user interface **116**, can allow the user **101** access to applications, hardware, data, or the like that is hosted on a network remote to the device

102. The web browser **120** can be used to request access to the functionality of the cloud **106**, such as through the network **104**, sometimes called the Internet.

(19) The network **104** is a series of routers, switches, hubs, or the like, connected to other networks, such as the cloud **106**, or user devices. The network **104** can be the medium through which the device **102** gains access to functionality the user **101** desires but may not have access to locally.

(20) The cloud **106**, as illustrated, includes a database **108**, applications **110**, hardware **112**, and network circuitry (illustrated as a firewall/router **114**). The database **108** includes data, such as a file, webpage, text, image, video, audio, a combination thereof, or the like. The database **108** can be accessed through permissions enforced by the firewall/router **114** or other network circuitry.

(21) The applications **110** include software functionality. The applications **110** can perform operations of software, which can include accessing data in the database **108**, functionality of the hardware **112**, or the like. The applications **110** can perform any operations that can be performed by a computer, cluster of computers or processors, or the like.

(22) The hardware **112** can include one or more computers, a cluster of computers, a data server, one or more graphics processing units (GPUs), field programmable gate arrays (FPGAs), application specific integrated circuits (ASICs), central processing units (CPUs), or the like. An application **110** can operate on the hardware **112**, such as to operate as a virtual machine.

(23) The firewall/router **114** can implement security controls for the cloud **106**. Security controls can be defined by an administrator. The security controls can define devices that are allowed access to resources (e.g., the database **108**, applications **110**, or hardware **112**) of the cloud **106** and the resources, or portions thereof, to which the devices are allowed access. The security controls can include password or other authentication information that is required to be provided before the device is allowed access to the resource.

(24) The firewall/router **114** can monitor traffic therethrough. The firewall/router **114** can implement a neural network (NN) configured to classify the traffic into normal activity classifications (e.g., file access, application access, hardware access) and unknown activity classification (e.g., potentially malicious activity). The NN can classify the traffic based on sampled traffic metadata. Sample traffic metadata is explained in more detail regarding FIG. 2.

(25) FIG. 2 illustrates, by way of example, a diagram of an embodiment of a system **200** for generating sampled network metadata **228**. The system **200**, as illustrated, includes a packet **202**, an analysis operation **226**, and a sample operation **232**. The packet **202** typically includes a header **204**, a payload **206**, and a trailer **208**. The header **204** generally indicates an source address **210** of a device that originated the packet **202**, a destination address **212** of a device to receive the payload **206** of the packet **202**, a protocol **214** to be used in transferring the packet **202** between devices, a number of packets **216** related to a total payload (of which the payload **206** is at least a part), and a packet number **218** that indicates, of the number of packets indicated by the number of packets **216**, which the packet **202** corresponds to. The payload **206** is at least a portion of the data to be transmitted to the destination device. The trailer **208** indicates an end of the packet **220** so that devices can determine where one packet ends and another packet begins. The trailer **208** can include a cyclic redundancy check (CRC) **222** that allows for error checking on the header, payload, trailer, or a combination thereof.

(26) The entire packet **202** is sometimes called deep packet information **224**. To perform security analysis (not to be confused with the analysis operation **226**) on the deep packet information **224** is prohibitively expensive in terms of compute resources, memory consumed, and network bandwidth consumed. Security analysis on the deep packet information **224** slows down data transmission, as the data transmitted is either stalled while analysis or performed or is copied and then transmitted before analysis is performed.

(27) The analysis operation **226** can generate metadata **228** based on the deep packet information **224**. Programs like Netflow and Internet Protocol version 6 can perform such analysis on the deep packet information **224**. The metadata **228** describes the data of the deep packet information **224**.

The metadata **228** can include capture of information, such as source address **210** or destination address **212**, protocol **2114**, source and destination transmission control protocol (TCP)/user datagram protocol (UDP) ports, service (associated application **110**), time/date stamps, data quantity, or a combination thereof. Examples of programs or protocols to generate the metadata **228** include NetFlow, Argus, App Flow, JStream, JFLOW, among others. The analysis operation **226** can be performed by a switch, packet broker, router, firewall, or the like. Note the packet **202** is transmitted over layer three of the OSI infrastructure.

(28) Application metadata **230**, in contrast to network data, is transmitted over layer 7 of the OSI architecture. Application metadata **230** can be generated in a manner similar to the packet metadata **228**, but based on application operation information (e.g., ports accessed, data requested, a device that is the subject of the data request, or the like).

(29) The sample operation **232** can reduce the totality of the metadata **228** to a subset of the metadata **228**. The sample operation **232** can be a uniform or non-uniform sampling. Sampling means keeping only a portion of the metadata **228**. Sampling can include retaining metadata of every other packet, every third packet, every fourth packet, every fifth packet, . . . every tenth packet, . . . every hundredth packet, . . . every thousandth packet, . . . etc., every two out of three packets, . . . or some other sampling rate therebetween. The metadata **228** retained after the sampling operation is the sampled metadata **234**. At this point, the deep packet information **224** has been reduced to metadata **228** and reduced again to sampled metadata **234**. This greatly reduces the data burden for traffic analysis.

(30) FIG. 3 illustrates, by way of example, a conceptual block diagram of a system **300** for training an NN **330** to classify network traffic based on only the sampled metadata **234**. The system **300**, as illustrated, includes the NN **330**. The NN **330** receives the sampled metadata **234** and the deep packet information **224** as input. The NN **330** determines a classification **336** based on the input. An actual classification **334** for the deep packet information **224** is compared to the NN classification **336** at operation **332**. The deep packet information **224** and the actual classification **334** of the deep packet information **224** are publicly available. The sampled metadata **234** can be determined using the analysis operation **226** and the sample operation **332** of FIG. 2. The deep packet information **224**, sampled metadata **234**, and actual classification **334** can form the training data set. Generally, the more training data used, the more accurate the NN **330** can operate after deployment. Initial parameters of the NN **330** can be guessed or set to a random value and adjusted through backpropagation.

(31) A difference between the actual classification **334** and the NN classification **336** can define a loss **338**. The loss **338** can be determined as a least absolute error, least squared error, hinge loss, logistic loss, or the like. The loss **338** can be backpropagated to the NN **330** to adjust parameters of the neurons of the NN **330**. The classification performed on the sampled metadata **234** can label the sampled metadata **234** with actions previously only detected based on inspection of the deep packet information **224**.

(32) The NN **330** can include bidirectional long short term memory (LSTM) gates, gated recurrent units (GRUs), or the like. Details of bidirectional LSTMs are described in more detail regarding FIG. 5.

(33) FIG. 4 illustrates, by way of example, a conceptual block diagram of an embodiment of a system **400** for using the NN **330** after training. The NN **440** represents the NN **330** after training. The NN **440** can classify network traffic based on only the sampled metadata **234**. The manner in which the NN **440** was trained, as discussed with FIG. 3, embeds knowledge of the deep packet information **224** associated with the sampled metadata **234** into the gates or units of the NN **440**. The NN **440**, however, does not require the deep packet information **224** to accurately make the classification **336**. The NN **440** can operate using only the sampled metadata **234** to generate the classification **336**.

(34) The classification **336** can include a network traffic action as the label for the sampled

metadata **234**. The classification can include device authentication, user authentication, querying a database, brute force attack, volumetric attack, accessing an application, transferring a file, streaming data, or the like. Examples of classifications that are typically determined based on the deep packet information **224** include device authentication, user authentication, querying a database, brute force attack, accessing an application, transferring a file, and streaming data. The sampled metadata **234** can be provided along with the NN classification **336** to an analyzer. The analyzer can determine whether to act based on the sampled metadata **234** and/or the classification **336**.

(35) FIG. 5 illustrates, by way of example, a conceptual block diagram of an embodiment of an analysis system **500**. The analysis system **500** as illustrated includes an analyzer **502** that receives the NN classification **336** and optionally the sampled metadata **334**. The analyzer **502** can include a software program, hardware such as electric or electronic components configured to perform operations of the analyzer **502**, a subject matter expert (SME), a combination thereof, or the like. The analyzer **502** can determine, based on the NN classification **336** and/or the sampled network metadata **334** if and what action **504** is to be performed to help mitigate the malware attack. The analyzer **502**, for example, can determine that the NN classification **336** indicates that a malware attack is being performed by a text message, email, an icon on a display (e.g., flashing, color encoded, annotated, or the like), a sound, or the like. The malware attack can be indicated by a variety of patterns in the NN classification **336** and/or sample network metadata **224**. For example, a data exfiltration following a successful brute force attack, an anomalous downloading or uploading of certain file types (e.g., a portable data format (pdf) download from websites where most downloads are of image files, or the like), a file upload from an unauthorized user, among many others.

(36) FIG. 6 illustrates, by way of example, a conceptual block diagram of an embodiment of an LSTM NN **600**. The LSTM NN **600** receives input **650A**, **650B**, **650C** and produces hidden feature values **660A**, **660B**, **660C** based on the input **650A-650C**. The input **650A-650C** in training can include the sampled metadata **234** and the deep packet information **224**, features thereof, or an embedded version thereof, such as during training. The input **650A-650C** during runtime can include just the sampled metadata **234**, features of the sampled metadata **234**, a word embedding of the sampled metadata **234**, or a combination thereof.

(37) The input **650A-650C** can be provided to forward encoding neurons **652A**, **652B**, **652C** and backward encoding neurons **654A**, **654B**, **654C**. The forward encoding neurons **652A-652C** can encode the neurons in order from least significant unit (input **650A**) to most significant unit (input **650C**). The forward encoding neuron **652A** encodes the input **650A** and provides the encoding to the forward encoding neuron **654A** and a combination operation **658A**. The forward encoding neuron **654A** encodes based on both the encoding from the forward encoding neuron **652A** and the input **650B** and provides the determined encoding to the next forward encoding neuron in the sequence and the combination operation **658B**. The process continues until a final forward encoding neuron **656A** which encodes based on an encoding of all prior input **650A**, **650B**, etc. and provides the determined encoding to just the combination operation **658C**.

(38) The backward encoding neurons **654A-654C** operate similar to the forward encoding neurons **652A-652C** but encode in reverse order through the input **650A-650C**. Thus, the input **650C** is encoded first by the backward encoding neuron **654C** and the encoding is provided to both the combination operation **658C** and a next backward encoding neuron. The backward encoding neurons **654A-654C** continue to operate on the input in reverse order until the backward encoding neuron **654A** encodes based on the encoding from the backward encoding neuron **654B** and the input **650A**.

(39) The combination operation **658A**, **658B**, **658C** can include an addition, multiplication, average, concatenation, or the like. The output **660A**, **660B**, **660C** is thus a combination of the encoding of the input **650A-650C** in the forward direction (from the forward encoding neurons

652A-652C) and the backward direction (from the backward encoding neurons 654A-654C). The output 660A-660C can then be used for classification. The bi-directional LSTM NN 600 is thus trained to embed deep packet information 224 knowledge into the sample metadata 234 for determining the classification 336.

(40) FIG. 7 illustrates, by way of example, a flow diagram of an embodiment of a method 700 for cloud security. The method 700 can be performed by one or more of the components of FIGS. 1-6, such as the firewall/router 114 or a different device of the network 104, the cloud 106, or a device coupled therebetween. The method 700 can include receiving (at processing circuitry of a device) sampled network metadata of a packet transmitted via a computer network, at operation 702; providing, by the processing circuitry, the sampled network metadata to a neural network (NN) trained on labeled sampled network metadata, at operation 704; and providing, based on only the sampled network metadata, a classification for the sampled network metadata via the trained neural network, at operation 706.

(41) The method 700 can further include, wherein the NN is trained further based on contents of the packet and the label is an actual classification associated with the contents of the packet and associated sampled network metadata. The method 700 can further include, wherein the actual classification is determined using deep packet inspection. The method 700 can further include, wherein the NN is a recurrent NN. The method 700 can further include, wherein the NN includes a bi-directional long short term memory (LSTM) NN.

(42) The method 700 can further include, wherein the sampled network metadata is of network traffic provided over layer three of the computer network. The method 700 can further include, wherein the actual classification includes one of a user authentication, a device authentication, a database query, file transfer, data streaming, or a malicious action.

(43) Electric or electronic components of the device 102, network 104, cloud 106, or other device or circuitry can include one or more transistors, resistors, capacitors, inductors, amplifiers, modulators, demodulators, antennas, radios, regulators, diodes, oscillators, multiplexers, logic gates (e.g., AND, OR, XOR, negate, or the like), buffers, caches, memories, GPUs, CPUs, FPGAs, ASICs, or the like.

(44) Artificial intelligence (AI) is a field concerned with developing decision-making systems to perform cognitive tasks that have traditionally required a living actor, such as a person. Neural networks (NNs) are computational structures that are loosely modeled on biological neurons. Generally, NNs encode information (e.g., data or decision making) via weighted connections (e.g., synapses) between nodes (e.g., neurons). Modern NNs are foundational to many AI applications, such as speech recognition.

(45) Many NNs are represented as matrices of weights that correspond to the modeled connections. NNs operate by accepting data into a set of input neurons that often have many outgoing connections to other neurons. At each traversal between neurons, the corresponding weight modifies the input and is tested against a threshold at the destination neuron. If the weighted value exceeds the threshold, the value is again weighted, or transformed through a nonlinear function, and transmitted to another neuron further down the NN graph—if the threshold is not exceeded then, generally, the value is not transmitted to a down-graph neuron and the synaptic connection remains inactive. The process of weighting and testing continues until an output neuron is reached; the pattern and values of the output neurons constituting the result of the ANN processing.

(46) The correct operation of most NNs relies on accurate weights. However, NN designers do not generally know which weights will work for a given application. NN designers typically choose a number of neuron layers or specific connections between layers including circular connections. A training process may be used to determine appropriate weights by selecting initial weights. In some examples, the initial weights may be randomly selected. Training data is fed into the NN and results are compared to an objective function that provides an indication of error. The error indication is a measure of how wrong the NN's result is compared to an expected result. This error

is then used to correct the weights. Over many iterations, the weights will collectively converge to encode the operational data into the NN. This process may be called an optimization of the objective function (e.g., a cost or loss function), whereby the cost or loss is minimized.

(47) A gradient descent technique is often used to perform the objective function optimization. A gradient (e.g., partial derivative) is computed with respect to layer parameters (e.g., aspects of the weight) to provide a direction, and possibly a degree, of correction, but does not result in a single correction to set the weight to a “correct” value. That is, via several iterations, the weight will move towards the “correct,” or operationally useful, value. In some implementations, the amount, or step size, of movement is fixed (e.g., the same from iteration to iteration). Small step sizes tend to take a long time to converge, whereas large step sizes may oscillate around the correct value or exhibit other undesirable behavior. Variable step sizes may be attempted to provide faster convergence without the downsides of large step sizes.

(48) Backpropagation is a technique whereby training data is fed forward through the NN—here “forward” means that the data starts at the input neurons and follows the directed graph of neuron connections until the output neurons are reached—and the objective function is applied backwards through the NN to correct the synapse weights. At each step in the backpropagation process, the result of the previous step is used to correct a weight. Thus, the result of the output neuron correction is applied to a neuron that connects to the output neuron, and so forth until the input neurons are reached. Backpropagation has become a popular technique to train a variety of NNs. Any well-known optimization algorithm for back propagation may be used, such as stochastic gradient descent (SGD), Adam, etc.

(49) FIG. 8 illustrates, by way of example, a block diagram of an embodiment of a machine **800** (e.g., a computer system) to implement one or more embodiments. The machine **800** can implement a technique for secure cloud application provision, such as the method **700**. The user device **102**, network **104**, cloud **106**, or a component thereof can include one or more of the components of the machine **800**. One example machine **800** (in the form of a computer), may include a processing unit **802**, memory **803**, removable storage **810**, and non-removable storage **812**. Although the example computing device is illustrated and described as machine **800**, the computing device may be in different forms in different embodiments. For example, the computing device may instead be a smartphone, a tablet, smartwatch, or other computing device including the same or similar elements as illustrated and described regarding FIG. 8. Devices such as smartphones, tablets, and smartwatches are generally collectively referred to as mobile devices. Further, although the various data storage elements are illustrated as part of the machine **800**, the storage may also or alternatively include cloud-based storage accessible via a network, such as the Internet.

(50) Memory **803** may include volatile memory **814** and non-volatile memory **808**. The machine **800** may include—or have access to a computing environment that includes—a variety of computer-readable media, such as volatile memory **814** and non-volatile memory **808**, removable storage **810** and non-removable storage **812**. Computer storage includes random access memory (RAM), read only memory (ROM), erasable programmable read-only memory (EPROM) & electrically erasable programmable read-only memory (EEPROM), flash memory or other memory technologies, compact disc read-only memory (CD ROM), Digital Versatile Disks (DVD) or other optical disk storage, magnetic cassettes, magnetic tape, magnetic disk storage or other magnetic storage devices capable of storing computer-readable instructions for execution to perform functions described herein.

(51) The machine **800** may include or have access to a computing environment that includes input **806**, output **804**, and a communication connection **816**. Output **804** may include a display device, such as a touchscreen, that also may serve as an input device. The input **806** may include one or more of a touchscreen, touchpad, mouse, keyboard, camera, one or more device-specific buttons, one or more sensors integrated within or coupled via, wired or wireless data connections to the

machine **800**, and other input devices. The computer may operate in a networked environment using a communication connection to connect to one or more remote computers, such as database servers, including cloud-based servers and storage. The remote computer may include a personal computer (PC), server, router, network PC, a peer device or other common network node, or the like. The communication connection may include a Local Area Network (LAN), a Wide Area Network (WAN), cellular, Institute of Electrical and Electronics Engineers (IEEE) 802.11 (Wi-Fi), Bluetooth, or other networks.

(52) Computer-readable instructions stored on a computer-readable storage device are executable by the processing unit **802** (sometimes called processing circuitry) of the machine **800**. A hard drive, CD-ROM, and RAM are some examples of articles including a non-transitory computer-readable medium such as a storage device. For example, a computer program **818** may be used to cause processing unit **802** to perform one or more methods or algorithms described herein.

ADDITIONAL NOTES AND EXAMPLES

(53) Example 1 includes a device comprising processing circuitry and a memory including instructions that when executed by the processing circuitry cause the processing circuitry to perform operations, the operations comprising receiving sampled network metadata of a packet transmitted via a computer network, providing the sampled network metadata to a neural network (NN) trained on labeled sampled network metadata, and providing, based on only the sampled network metadata, a classification for the sampled network metadata via the trained neural network.

(54) In Example 2, Example 1 can further include, wherein the NN is trained further based on contents of the packet and the label is an actual classification associated with the contents of the packet and associated sampled network metadata.

(55) in Example 3, Example 2 can further include, wherein the actual classification is determined using deep packet inspection.

(56) in Example 4, at least one of Examples 1-3 can further include, wherein the NN is a recurrent NN.

(57) In Example 5, at least one of Examples 1-4 can further include, wherein the NN includes a bi-directional long short term memory (LSTM) NN.

(58) in Example 6, at least one of Examples 1-5 can further include, wherein the sampled network metadata is of network traffic provided over layer three of the computer network.

(59) In Example 7, at least one of Examples 1-6 can further include, wherein the actual classification includes one of a user authentication, a device authentication, a database query, file transfer, data streaming, or a malicious action.

(60) In Example 8, at least one of Examples 1-7 can further include, wherein the device is a router, switch, firewall, or client device.

(61) Example 9 includes a method including operations or (non-transitory) machine-readable medium including instructions that, when executed by a machine cause the machine to perform the operations, the operations comprising receiving (at processing circuitry of a device) sampled network metadata of a packet transmitted via a computer network, providing (by the processing circuitry) the sampled network metadata to a neural network (NN) trained on labeled sampled network metadata, and providing, based on only the sampled network metadata, a classification for the sampled network metadata via the trained neural network.

(62) In Example 10, Example 9 can further include, wherein the NN is trained further based on contents of the packet and the label is an actual classification associated with the contents of the packet and associated sampled network metadata.

(63) in Example 11, at least one of Examples 9-10 can further include, wherein the actual classification is determined using deep packet inspection.

(64) In Example 12, at least one of Examples 9-11 can further include, wherein the NN is a recurrent NN.

(65) in Example 13, at least one of Examples 9-12 can further include, wherein the NN includes a bi-directional long short term memory (LSTM) NN.

(66) In Example 14, at least one of Examples 9-13 can further include, wherein the sampled network metadata is of network traffic provided over layer three of the computer network.

(67) In Example 15, at least one of Examples 9-14 can further include, wherein the actual classification includes one of a user authentication, a device authentication, a database query, file transfer, data streaming, or a malicious action.

(68) Although a few embodiments have been described in detail above, other modifications are possible. For example, the logic flows depicted in the figures do not require the order shown, or sequential order, to achieve desirable results. Other steps may be provided, or steps may be eliminated, from the described flows, and other components may be added to, or removed from, the described systems. Other embodiments may be within the scope of the following claims.

Claims

1. A device comprising: processing circuitry; a memory including instructions that when executed by the processing circuitry cause the processing circuitry to perform operations, the operations comprising: performing deep packet inspection on deep network information of network traffic, including a packet transmitted via a computer network, resulting in network metadata; sampling the network metadata to include data of a subset of the packets associated with the network metadata resulting in sampled network metadata; providing the sampled network metadata to a recurrent neural network (RNN) trained (i) on labeled sampled network metadata and the deep network information and (ii) to generate a classification based on only the sampled network metadata, the classification indicating whether the network traffic associated with the sampled network metadata is malicious; generating, by an analyzer and based on the classification for the sampled network metadata, an action, the action indicating an operation to mitigate malicious traffic associated with the sampled network metadata; and performing the action.

2. The device of claim 1, wherein the RNN is trained further based on contents of the packet and the label is an actual classification associated with the contents of the packet and associated sampled network metadata.

3. The device of claim 2, wherein the actual classification is determined using the deep packet inspection.

4. The device of claim 2, wherein the actual classification includes one of a user authentication, a device authentication, a database query, file transfer, data streaming, or a malicious action.

5. The device of claim 1, wherein the RNN includes a bi-directional long short term memory (LSTM) NN.

6. The device of claim 1, wherein the sampled network metadata is of network traffic provided over layer three of the computer network.

7. The device of claim 1, wherein the device is a router, switch, firewall, or client device.

8. A method comprising performing, by processing circuitry, deep packet inspection on deep network information of network traffic, including a packet transmitted via a computer network, resulting in network metadata; sampling the network metadata to include data of a subset of the packets associated with the network metadata resulting in the sampled network metadata; providing, by the processing circuitry, the sampled network metadata to a recurrent neural network (RNN) trained (i) on labeled sampled network metadata and the deep network information and (ii) to generate a classification based on only the sampled network metadata, the classification indicating whether the network traffic associated with the sampled network metadata is malicious; generating, by an analyzer and based on the classification for the sampled network metadata, an action, the action indicating an operation to mitigate malicious traffic associated with the sampled network metadata; and performing the action.

9. The method of claim 8, wherein the RNN is trained further based on contents of the packet and the label is an actual classification associated with the contents of the packet and associated sampled network metadata.

10. The method of claim 9, wherein the actual classification is determined using the deep packet inspection.

11. The method of claim 9, wherein the actual classification includes one of a user authentication, a device authentication, a database query, file transfer, data streaming, or a malicious action.

12. The method of claim 8, wherein the RNN includes a bi-directional long short term memory (LSTM) NN.

13. The method of claim 8, wherein the sampled network metadata is of network traffic provided over layer three of the computer network.

14. A non-transitory machine-readable medium including instructions that, when executed by a machine, cause the machine to perform operations comprising: performing deep packet inspection on deep network information of network traffic, including a packet transmitted via a computer network, resulting in network metadata; sampling the network metadata to include data of a subset of the packets associated with the network metadata resulting in the sampled network metadata; providing the sampled network metadata to a recurrent neural network (RNN) trained (i) on labeled sampled network metadata and (ii) to generate a classification based on only the sampled network metadata, the classification indicating whether the network traffic associated with the sampled network metadata is malicious; generating, by an analyzer and based on the classification for the sampled network metadata, an action, the action indicating an operation to mitigate malicious traffic associated with the sampled network metadata; and performing the action.

15. The non-transitory machine-readable medium of claim 14, wherein the RNN is trained further based on contents of the packet and the label is an actual classification associated with the contents of the packet and associated sampled network metadata.

16. The non-transitory machine-readable medium of claim 15, wherein the actual classification is determined using the deep packet inspection.

17. The non-transitory machine-readable medium of claim 14, wherein the RNN includes a bi-directional long short term memory (LSTM) NN.
